

Group # : 7

Lab 2.1 Answer

```
#include <iostream>

//Original code lack of termination, causing infinite recursion.A update need to be made so
that we can get the correct n^2

using namespace std;

// Function for n^2

int powerOfTwo(int n) {

    // Base case: if n is 0, return 0

    if (n == 0) {

        return 0;

    }

    // Base case: if n is 1,then return 1

    if (n == 1) {

        return 1;

    }

    // Recursive case: return n + powerOfTwo(n - 1) + (n - 1)

    return n + powerOfTwo(n - 1) + (n - 1);

}

int main() {

    //Prompt the output for square of 3,0,1

    cout <<powerOfTwo(3) << endl;

    cout <<powerOfTwo(0) << endl;

    cout <<powerOfTwo(1) << endl;

    return 0;

}
```

```
9
0
1

-----
Process exited after 0.06774 seconds with return value 0
Press any key to continue . . . |
```

Lab 2.2 Answer

```
#include <iostream>

using namespace std;

int sumUpTo(int n) {
    if (n<=0) //Base case that return 0 when n =0
        return 0;
    else
        return n+sumUpTo(n- 1); //Recursive case :Adding number down to 1
}

int main() {
    int n;

    cout << "Enter a positive integer: ";

    cin >> n;

    //prompt the sum from 1 to the n enter

    cout <<" The sum of all integers from 1 up to " << n << " is " <<sumUpTo(n)<< endl;

    return 0;
}
```

```

Enter a positive integer: 5
The sum of all integers from 1 up to 5 is 15

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Process exited after 2.89 seconds with return value 0
Press any key to continue . . . |

```

Total Score: /15 (Each student rates themselves and each of their peers individually)
 Submission: Each student submits an individual assessment of their own contributions and those of their group members based on contributions and teamwork.

Evaluator/Criteria	Technical Skills(/4)	Idea Contribution(/4)	Engagement & Seriousness (/4)	Collaboration & Team Spirit (/3)	Total(/15)
JOANNE CHING YIN XUAN	4	4	4	3	15
CHOH JING YI	4	4	4	3	15
MUHAMMAD AFIQ DANISH BIN MOHD HAZNI	4	4	4	3	15
TAN SHEAU JUN	4	4	4	3	15